Guillermo López

Senior FullStack & AI Engineer

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Summary

Senior FullStack and AI engineer with 6+ years building distributed systems in TypeScript/Node (NestJS, Kafka, PostgreSQL, MongoDB) on AWS. CTO at Vlue Logistics, where I lead a team of 10 engineers and own a platform serving 50K drivers and riders and processing 80M requests a day. Over the last two years I've shipped production LLM systems: a multi-agent healthcare platform (ZentrumCare), a RAG engine for engineering compliance, a mindfulness app with AI-generated meditation sessions, open-source MCP tooling, and blobot, an open-source desktop app that runs Claude Code, Codex and Cursor as one team on a repo. Maintainer of Morphicons (2K GitHub stars, 20K+ weekly npm downloads).

Experience

CTO & Backend Lead, Vlue Logistics LLC – Miami, FL (Remote) Feb 2023 – present

- Promoted from Backend Lead to CTO. Own the full platform behind 50K drivers and riders: 4 native mobile apps (iOS/Android), back-office, support dashboard, backend, AWS infrastructure and public website.
- Scaled the backend to 80M requests a day across HTTP traffic, real-time driver positions and logs, on NestJS, Kafka, MQTT, Redis, Firestore and ClickHouse.
- Lead a team of 10 engineers across mobile, backend and DevOps; set up code review, CI/CD on GitHub Actions and Scrum workflows, and take part in executive-level strategic decisions.
- Migrated the legacy Node API to a clean architecture (DDD, CQRS, Hexagonal, Repository pattern, polyglot persistence) with zero downtime while the team kept shipping features.
- Designed the real-time tracking and driver positioning system; integrated Here Maps, Stripe, AppsFlyer and Meta Conversions API.
- Modernized the AWS infrastructure progressively (Docker, Kubernetes, Terraform) without disrupting development or customer experience.

Full Stack & AI Engineer, Independent – Remote (part-time, alongside full-time roles) 2022 – present

- **blobot** (2026, in progress) | Co-creator of an open-source desktop app that runs the coding agents already installed on a machine (Claude Code, Codex, Cursor) as one team on the same repo: each agent works in its own git worktree and branch, can hand part of a task to a teammate, and can run isolated in a Docker sandbox. Electron, React 19, TypeScript and SQLite on top of the Agent Client Protocol, with five runtime adapters tested against the real CLIs and per-agent permission levels.
- **ZentrumCare** (2026) | Built from scratch a HIPAA-compliant multi-agent platform that takes US families from first contact to a booked pediatric appointment in under 48 hours. 9+ specialized agents cover chat intake, voice confirmation of each child's condition (ElevenLabs), provider matching against the internal network with NPI Registry and Google Places fallback, sales outreach, customer support and booking assistance. TypeScript monorepo (6 apps, 16 packages) on Next.js, tRPC, PostgreSQL/Drizzle, Trigger.dev and Vercel AI SDK; every agent run traced in PostHog (LLM analytics, session replay, error tracking, feature flags); ~60 ADRs.
- **phimn.ai** | Internal AI platform for a structural engineering firm that automates code-compliance calculations: RAG over normative documents combined with LLM reasoning, a web-search subagent and Inngest-based orchestration.
- **mongodb-mcp-vercel** | Open-source read-only MongoDB Model Context Protocol (MCP) server for serverless deployment on Vercel, giving LLM agents governed access to data.
- **Numa365** | React Native mindfulness | Horoscope app with AI-generated meditation sessions delivered end to end as an independent engineer.

Backend & DevOps Engineer, InformageStudio, contracted to Mutter Ventures – Spain (Remote) Oct 2022 – Feb 2023

- Ran AWS infrastructure and maintained backend codebases for a startup incubator/VC firm and a software consultancy.
- Designed backend solutions for new product features under tight delivery cycles.

Backend Lead, Contapp Digital – Dominican Republic (Remote) 2020 – Oct 2022

- Led a backend team building one of the first automated accounting platforms in the Dominican Republic, used by businesses and accountants on web, iOS and Android.
- Shipped the platform's backend and some early AI features: receipt and invoice OCR, AI analysis of a company's accounting data, automatic chart generation and invoice creation.
- Migrated the Express/TypeScript monolith to NestJS microservices (RabbitMQ, Redis) with Hexagonal Architecture, DDD, CQRS and Jest test coverage, so services could be built, tested and deployed independently.

Full Stack Developer, Freelance – Cuba (Remote) 2019 – 2020

- Delivered Shopify stores, CMS-based websites and small internal tools for local businesses and foreign clients on the MERN stack (Shopify, WordPress, Contentful, BigCommerce).

Skills

Languages: TypeScript, JavaScript, Python (entry level)

Backend: Node.js, NestJS, Express, tRPC, Next.js

AI & Agents: LLM applications, multi-agent systems, agent orchestration, Model Context Protocol (MCP) servers, RAG, tool calling, prompt engineering, AI gateways (Vercel AI Gateway, OpenRouter, TanStack), Vercel AI SDK, TanStack AI SDK, Trigger.dev, Inngest, ElevenLabs, LLM observability (PostHog)

AI tooling: Claude Code, Codex, Cursor...

Messaging & Cache: Kafka, RabbitMQ, MQTT, Redis

Databases: PostgreSQL (Drizzle, Prisma), MongoDB, Firestore, ClickHouse

Frontend & Mobile: React, Next.js, React Native, TanStack, Vite, Tailwind CSS, shadcn/ui

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Vercel

Architecture: DDD, CQRS, Hexagonal, Microservices, Event-driven, Monorepos

Methodologies: Agile/Scrum, technical leadership, code review, mentoring, ADRs

Languages (spoken): Spanish (native), English (professional working proficiency)

Education

Universidad de Camagüey, BS in Civil Engineering – Camagüey, Cuba Sept 2017 – July 2021

- Graduated with honors. Student Research Award (2019).

Universidad Adolfo Ibáñez, Postgraduate Diploma in Communication & Relational Negotiation – Chile (Online) Jan 2020 – Dec 2021

Open Source

- blobot - co-created desktop app that runs Claude Code, Codex and Cursor as one team on a repo, each agent in its own git worktree. Work in progress.
- morphicons - Morph any SVG icon into any other with math, no dependencies. ~2K GitHub stars, 20K+ weekly npm downloads, community ports and integrations.
- translate-kit - npm package that automates i18n setup and maintenance in Next.js with AI.
- nextjs-stripe-store - ready-to-use Next.js + Stripe template.
- Contributor to trigger.dev and blobatar.dev.